

M O D U L E 0 1

Introduction to Information Security

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Overview

This report covers the fundamentals of Information Security (InfoSec), including core principles, key roles, and essential real-world security concepts. Security is not just about tools — it is a complete system involving technical controls, human behavior, and organizational policies.



The CIA Triad

The CIA Triad is the foundation of all information security decisions:

Confidentiality	Integrity	Availability
Only authorized users can access data	Data should not be modified without permission	Data should be accessible when needed



Structure of InfoSec

Strong information security is built on three interconnected pillars. No single component is sufficient on its own.



People

- Employees and end users
- Security awareness training
- Human behavior and decision-making



Process

- Security policies and procedures

- Incident response planning
- Compliance and governance frameworks

Technology

- Firewalls and network controls
- IDS/IPS systems
- Encryption and access management

Strong Security = People + Process + Technology

Roles in Cybersecurity

Role	Responsibility
CISO	Overall security strategy & leadership
Security Analyst	Monitor & analyze threats
Penetration Tester	Find & report vulnerabilities
Incident Responder	Handle and contain active attacks

Network Security Basics

Key Network Controls

- Firewalls — Filter and control incoming/outgoing traffic
- IDS/IPS — Detect and prevent network-based attacks
- VPN — Secure, encrypted communication tunnels

Application Security

Core Practices

- Secure coding practices from the ground up

- Prevent common vulnerabilities: SQL Injection (SQLi), Cross-Site Scripting (XSS)
- Continuous testing, monitoring, and patching

Cloud Security

Key Principles

- Shared Responsibility Model — cloud provider vs customer duties
- IAM (Identity and Access Management) — control who can do what
- Encryption at rest and in transit, plus compliance requirements

Common Threats

Threat	Description
DDoS Attack	Overwhelm a system/network to make it unavailable
Ransomware	Encrypt victim data and demand payment for decryption
Social Engineering	Manipulate people into revealing sensitive information
Insider Threats	Malicious or negligent actions by internal users
APTs	Advanced Persistent Threats — long-term, targeted attacks by sophisticated actors

What I Learned

- ✓ Security is not just tools — it is also about people and processes working together.
- ✓ Human error remains one of the biggest and most underestimated security risks.
- ✓ Security must be built in from the beginning — Security by Design, not as an afterthought.

Real-World Insight

From this module, I understood that cybersecurity is a complete ecosystem. It is not sufficient to deploy technology alone — the entire system must function as one:

Dimension	Examples
Technical Controls	Firewalls, IDS/IPS, Encryption
Human Behavior	Awareness training, reducing error
Organizational Policies	Incident response, compliance, governance

Source Material: *HTB Academy – Introduction to Information Security | StationX | OpenSourceForU | MindMajix*
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